# MITOSYNC: PROGRAMMABLE DIVISION SUPERVISION FOR DIVISION-AWARE CELL TRACKING

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## ABSTRACT

Sparse sampling and limited annotations make mother–daughter association difficult. MitoSync combines joint daughter scoring with programmable division supervision. The tracker jointly decodes link scores and exchange-invariant triplet scores. Its generation pipeline programs divisions, schedules births and aligns labels with detected cells. On two retrospectively evaluated Cell Tracking Challenge sequences, sparse division F1 exceeds a Trackastra-based graph reference by 5.25 and 7.93 percentage points. On 36 external LFCT fields, the frozen real-only joint model exceeds this reference by 4.05 sparse event F1 points. Synthetic data add 0.79–1.70 points in three original full-tracker settings, but matched controls do not establish a consistent advantage over direct painting or real-only training. Joint scoring and decoding improve sparse tracking; rendered supervision has setting-dependent benefits.

***Index Terms*—** cell division, cell tracking, synthetic supervision, video diffusion, lineage

## 1. INTRODUCTION

Missed divisions and false daughter assignments corrupt lineages. Divisions are rare in tracker training data [1, 2]: fewer than 7 per 1000 cell-frames in our PhC-C2DL-PSC (PSC) training split. Sparse sampling separates mother and daughter observations in time.

Synthetic videos supply examples, but programmed divisions may appear late or be absent. MitoSync combines exchange-invariant daughter evidence with Trackastra link scores. Joint decoding makes divisions and continuations compete for cells. A programmable pipeline supplies supervision (Fig. 1). Controls separate association, appearance, label and sampling effects.

## 2. RELATED WORK

Cell videos can be simulated [3–5] or generated by learned models [6–9]. SynCellFactory (SCF) [10] combines fitted statistics, ControlNet and label refinement; ASIST [11] trains segmentation and tracking with simulations. LapTrack [12], MAGIK [2] and Trackastra [1] link detected cells. Zyss et al. [13] classify mother–two-daughter triplets using three-frame image and geometry features, select consistent divisions, then optimize tracks. We instead couple an exchange-invariant representation to Trackastra through a child-set utility that jointly selects competing divisions and continuations.

## 3. MITOSYNC

### 3.1. M1: programming cell divisions

M1 uses training-fitted statistics for lognormal cycles, growth, rounding and motion to generate 17-frame identity masks $L_{0:T}$ and lineage links. Initial ages follow the distribution in an exponentially growing population [14], avoiding synchronized divisions. The mother long axis defines the preferred division axis [15]. Division-axis deviation $\psi$ and daughter separation $d$ follow

$$p(\psi) \propto \exp\{\kappa \cos(2\psi)\}, \qquad d(k) = d_\infty - (d_\infty - d_0)e^{-k/\tau}. \quad (1)$$

Here $k$ counts frames after division. Training contact statistics guide placement, non-overlap and cell entry.

### 3.2. M2: phase scheduling and video generation

We adapt Wan2.1-VACE-1.3B [16, 17] using rank-32 low-rank adaptation (LoRA) [18] on its 15 VACE blocks. A latent flow-matching loss updates only LoRA parameters, using 113 PSC or 90 Fluo-N2DL-HeLa (Fluo) training windows. Controls encode instance boundaries, separation lines and distance transforms; a real training frame supplies appearance. Generation uses 30 steps and guidance scale 5 (Fig. 2).

The causal variational autoencoder (VAE) encodes frame 0 alone, then groups of four frames. Training calibration selects two birth phases $\phi(t_b) = (t_b - 1) \bmod 4$ with the highest on-time detection rates: $K^* = \{1, 2\}$ for PSC and $\{0, 2\}$ for Fluo. M2 shifts births to the nearest selected phase.

### 3.3. M3: aligning labels with generated frames

M3 matches frozen Cellpose-SAM [19] detections one-to-one with programmed masks at intersection over union (IoU)$\geq .5$. For a programmed birth $t_b$, the detected birth is

$$t^* = \min\{t \geq t_b : \text{both daughters match at } t \text{ and } t+1\}. \quad (2)$$

M3 keeps the birth when $t^* = t_b$ and re-times it to $t^*$ when delayed. If no match exists, it removes the split and assigns the daughter regions to the mother identity. This checks detector consistency; it does not independently establish ancestry. On PSC, M3 keeps 1085 of 6047 divisions on time (18%), re-times 2435 and drops 2527. The same rule detects 63% of real training divisions on time.

Acceptance requires 80% matched divisions and PSC instance F1 $\geq .6986$ (.05 below the calibration median). After two retries,

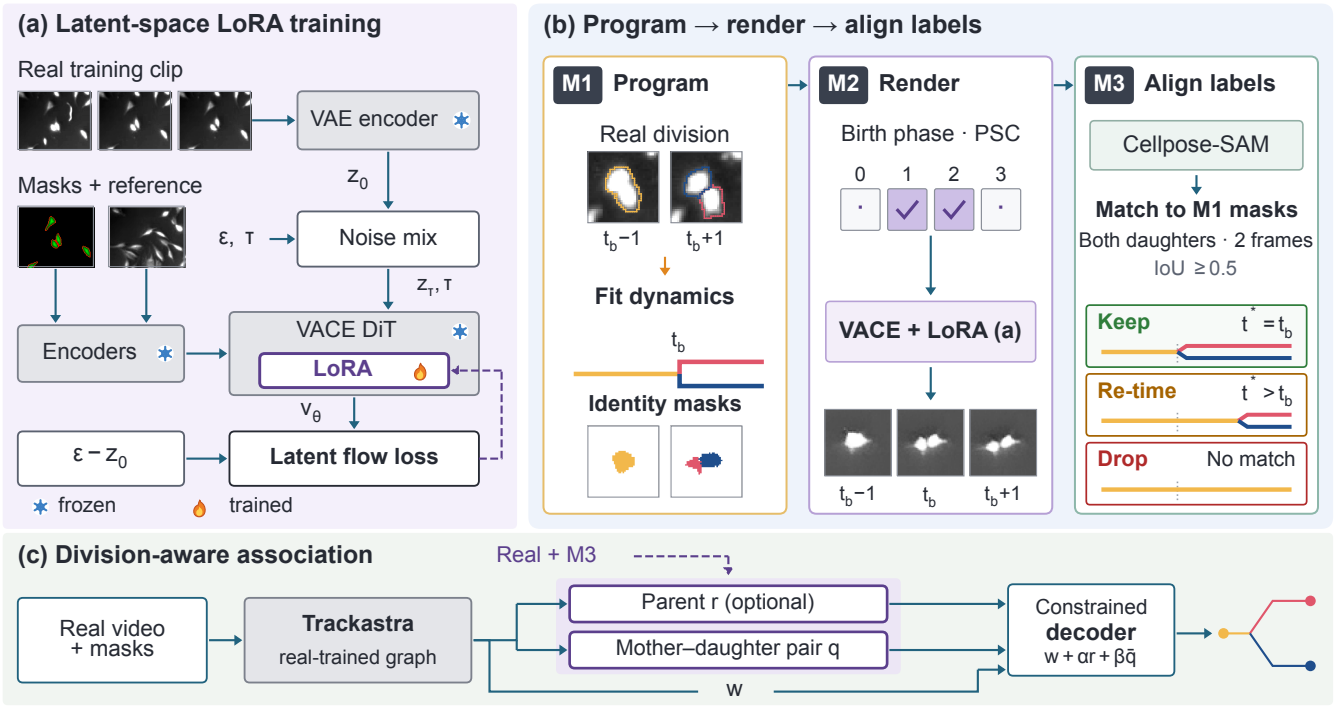



**Fig. 1**. MitoSync: (a) latent LoRA adaptation; (b) division programming, rendering and label alignment; (c) joint association. Dashed arrows carry training inputs/gradients; the direct path carries Trackastra score $w$. Yellow/pink/blue denote mother/daughters; $\tau$ is diffusion time.

fallback retains the highest-match-rate attempt passing the instance gate (176/191 PSC clips). Retained events satisfy Eq. (2).

### 3.4. MitoSync tracker

**Candidate links and decoding.** Real-trained Trackastra [1] scores consecutive links $p \to c$ with $w_{pc}$. Native-rate training adds recombined-cell frames; sparse-rate training adds eight subsampled views of the training split. Scores $r_{pc}$ and $q_p(a, b)$ refine links and mother–daughter triplets.

Each parent $p$ selects a set $S$ of at most three children; each child has at most one parent. Nonempty sets have utility

$$U_p(S) = \sum_{c \in S} \left(w_{pc} + \alpha r_{pc}\right) + \beta\, \bar{q}_p(S) + .75\,(2 - |S|). \quad (3)$$

Empty-set utility is 0; integer programming maximizes total utility. Setting $\alpha = \beta = 0$ gives the Trackastra graph reference in Table 1. For up to two children, this is the Trackastra integer linear programming (ILP) objective with deepcell_gt costs (.75). When learned scores are used, a parent with children in the reference must retain at least one child. The three-child limit accommodates annotations, not three-way mitosis.

**Mother–daughter pair head.** A frozen, real-trained ResNet-18 division classifier encodes three-frame cell crops as 256-dimensional vectors, projected to 32 dimensions. Boundary padding is flagged. A multilayer perceptron (MLP) combines the mother vector, daughter mean, absolute difference and product, and comparisons between mother and mean daughter features. It also uses 25 geometry/context values for area balance, separation, motion, density, frame gap and boundaries. The clipped logit $q_p(a, b) \in [-4, 4]$ is invariant to daughter swapping. Training uses balanced binary cross-entropy with hard negatives from the four nearest candidate children. In Eq. (3), $\bar{q}_p(S)$ averages $q_p$ over pairs in $S$; it is 0 when $|S| < 2$.

Validation selects $\beta = .125$ for PSC at 1× and .25 at 8× from $\{0, .125, .25\}$. Fluo at 8× uses the PSC weight; its native validation split, containing five divisions, selects .5 from $\{0, .125, .25, .5\}$.

**Optional parent head.** An MLP selects among the ten nearest parents and entry using image features and geometry (9 values at 1×, 15 at 8×). Clipped parent-versus-entry log odds give $r_{pc}$. PSC validation selects $\alpha = .25$ for both datasets and rates. Native training uses gap 1; sparse training balances gaps 1/2/4/8. Setting $\alpha = 0$ gives the simpler pair-only model.

**Training.** Both heads use 6000 AdamW updates with batch size 64; we evaluate final checkpoints. R uses real data; RG mixes real/synthetic data equally. Pair batches balance gaps 1/2/4/8 and positive/negative triplets. Five seeds vary initialization, sampling and Trackastra checkpoints; corpora and encoders remain fixed.

## 4. EXPERIMENTS AND RESULTS

### 4.1. Experimental setup

**Data.** PSC and Fluo each provide two annotated sequences [20, 21]. All components use video 02 frames 0–254 (PSC) and 0–77 (Fluo) for training, and the remaining 45/14 frames for parameter selection. Evaluation uses video 01: PSC (300 frames, 569 divisions) and Fluo (92 frames, 94 divisions). CTC methods share silver-truth masks.

MitoSync produces 191 accepted PSC clips and 200 Fluo clips. Their tracking labels contain 3520 and 3554 matched binary divisions, versus 260 and 202 real training divisions. Fluo counts include mothers ending at frame 0. Excluding them for classification context gives the earlier 3424/199 counts, without changing the corpus. Generation uses 584 PSC and 352 Fluo attempts including retries.

**Table 1**. Video 01 F1 (BC(0)) and complete tracks (CT), mean±SD; bold marks maxima. Greedy: released linker; Graph: $\alpha = \beta = 0$; Full: both heads; Pair: no parent scores. Detections are shared; greedy may omit isolated cells.

| | PSC 1× | | PSC 8× | | Fluo 1× | | Fluo 8× | |
|---|---|---|---|---|---|---|---|---|
| Method | F1 | CT | F1 | CT | F1 | CT | F1 | CT |
| LapTrack | .730 | .556 | .589 | .518 | .856 | .764 | .760 | .683 |
| MAGIK | .770±.005 | .536±.007 | .454±.019 | .377±.008 | .710±.019 | .608±.019 | .616±.035 | .512±.024 |
| + MitoSync | .797±.010 | .571±.011 | .472±.026 | .392±.014 | .729±.063 | .642±.036 | .500±.070 | .456±.050 |
| Trackastra greedy | .837±.007 | .651±.010 | .696±.007 | .592±.007 | .943±.009 | .861±.015 | .854±.024 | .787±.021 |
| Trackastra graph | .857±.008 | .660±.008 | .696±.007 | .594±.010 | .947±.010 | .871±.017 | .830±.038 | .767±.032 |
| Full (real only) | .862±.006 | .671±.006 | .741±.003 | .669±.004 | **.954**±.003 | **.884**±.005 | .899±.008 | .839±.005 |
| Full + SCF | .857±.009 | .665±.009 | .731±.005 | .660±.004 | .946±.007 | .880±.005 | .891±.006 | .831±.003 |
| Full + MitoSync | .879±.003 | .679±.002 | .749±.005 | **.674**±.004 | .950±.002 | **.884**±.006 | .909±.006 | .846±.003 |
| Pair (real only) | .874±.007 | .670±.009 | .749±.004 | .656±.007 | **.954**±.003 | **.884**±.005 | .909±.007 | .848±.004 |
| Pair + MitoSync | **.888**±.006 | **.682**±.004 | **.755**±.003 | .667±.004 | .950±.002 | .880±.005 | **.913**±.007 | **.854**±.004 |

**Table 2**. Component F1 over five seeds; bold marks maxima. No pair: no daughter scores; Geometry pair: zeroed image embeddings; Full: both heads. R/RG: real/real+synthetic.

| | | PSC 01 | | Fluo 01 | |
|---|---|---|---|---|---|
| Stage | Data | 1× | 8× | 1× | 8× |
| No pair | R | .838±.008 | .693±.006 | .909±.011 | .820±.015 |
| | RG | .866±.006 | .701±.009 | .928±.006 | .859±.015 |
| Geometry pair | R | .846±.006 | .724±.004 | .950±.003 | .886±.011 |
| | RG | .872±.004 | .737±.006 | .950±.002 | .898±.007 |
| Full | R | .862±.006 | .741±.003 | **.954**±.003 | .899±.008 |
| | RG | **.879**±.003 | **.749**±.005 | .950±.002 | **.909**±.006 |

**Evaluation.** Division F1 measures branching correctness with zero temporal tolerance (BC(0)) [20, 22]; CT measures complete-track recovery [20]. We evaluate native frames (1×) and every eighth frame (8×). Sparse ground truth retains observed tracks and parent links; a division requires the mother and at least two daughters. Eight correlated sampling offsets are averaged per seed. Five-seed comparisons use paired 95% $t$ intervals (four degrees of freedom) and Holm correction where stated. Differences are in percentage points. LFCT contrasts, identity audit and uncertainty masking are post hoc follow-ups. Video 01 informed earlier model development and the decision to reselect the Fluo native pair weight on validation data. The evaluation is therefore retrospective.

**Baselines.** LapTrack [12] uses parameters fitted on all of video 02. We train MAGIK [2] on geometry inputs, with and without MitoSync data. SCF [10] is retrained using its released code and schedule, with moment-matched training motion statistics. SCF retains simulated-position mask correction but uses generic Cellpose-SAM instead of fine-tuned Cellpose (199/200 programs per dataset). Table 1 uses the output of each pipeline; later controls match event budgets. LapTrack is deterministic; MAGIK uses three seeds and other trackers five.

### 4.2. Comparison with other trackers

Under sparse sampling (Table 1), full RG exceeds the Trackastra graph by 5.25 F1 points on PSC (95% interval [4.67,5.82]) and 7.93 on Fluo ([3.52,12.34]). It also exceeds the official greedy linker; comparisons include learned scores and continuity constraints. Sparse CT gains over graph are 7.96/7.92 points on PSC/Fluo. At 1×, the F1 gain over the graph is clear only on PSC; the Fluo F1 and CT intervals include zero.

### 4.3. Effects of tracking components and training data

Adding joint daughter scores produces the largest architectural gain (Table 2). Retraining with zeroed embeddings isolates image information. In RG, image features add 0.64–1.20 F1 points on PSC and sparse Fluo (eight-test Holm over R/RG, $p \leq .015$); native Fluo shows no clear effect. Without parent scores, RG image features add 3.16/1.83 sparse points on PSC/Fluo (separate eight-test Holm $p < .01$). Parent removal does not significantly change RG F1 after correction. Real-only pair models match full RG in rounded sparse F1 (Table 1). Matched and transfer tests use pair-only decoding.

With the full architecture fixed, synthetic data add 1.70 native PSC F1 points, 0.79 sparse PSC points and 0.99 sparse Fluo points; all paired intervals exclude zero. Sparse CT gains from synthetic data are 0.43 points on PSC and 0.76 on Fluo. On native Fluo, R leads RG by 0.39 F1 points ([0.12,0.65]).

### 4.4. Controls for synthetic supervision

**Labels and appearance.** Correcting labels on the same rendered images (Raw to RG, Table 3) adds 0.94/0.35 PSC F1 points at 1×/8×. Paired intervals exclude zero on PSC but include zero on Fluo. All RG–PM intervals include zero; rendered appearance has no established advantage over painting. PM uses M3-corrected labels.

**Direct-program control.** Painting all 200 M1 programs with original labels bypasses generation and M3 (Table 3, lower block). Intensity, texture and halos are fitted on real training frames. Both arms share schedules, 50:50 real/synthetic data, 6000 updates and five seeds. Events and negative-pool sizes are matched per gap, with one positive view per event. Rendering plus M3 adds 0.44 native PSC F1 points ([0.27,0.62]; four-test Holm $p = .009$); other intervals include zero. Neither arm differs significantly from real-only under separate eight-test Holm correction.

**Matched generator and tracker comparisons.** With matched events, negative pools and updates, MitoSync exceeds SCF by 1.19/2.37 sparse PSC/Fluo F1 points (four-test Holm $p < .002$). Native gains (+0.44/+0.20) are not significant. Generation costs remain unmatched. MAGIK loses 11.6 sparse Fluo points with uniform window sampling (Table 1). With balanced source and gap sampling (18,750 updates per arm), real-only/MitoSync-augmented Fluo F1

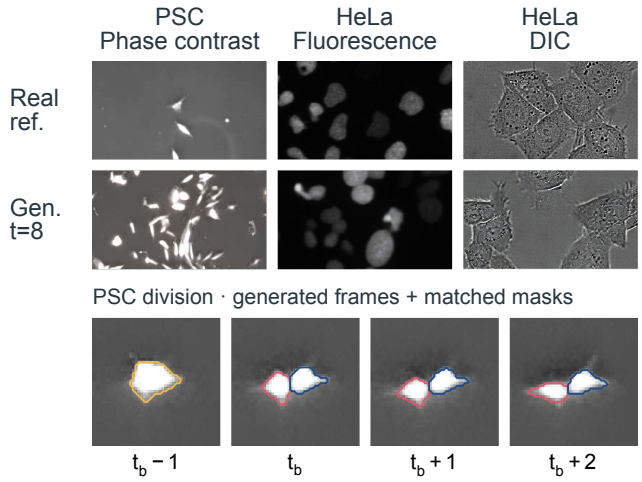



**Fig. 2**. Real references and generated images; below, consecutive $48 \times 48$-pixel PSC crops. DIC (differential interference contrast) is qualitative; tracking uses PSC and Fluo.

**Table 3**. Five-seed division F1; bold marks block maxima. Raw: rendered images/program labels; RG/PM: rendered/painted images with corrected labels. Lower block: original-label painting versus rendering/M3; matched events, pair-only decoding.

| | PSC 01 | | Fluo 01 | |
|---|---|---|---|---|
| Data | 1× | 8× | 1× | 8× |
| Full tracker; original sampling | | | | |
| Raw | .869±.006 | .745±.005 | .948±.005 | .908±.006 |
| RG | .879±.003 | .749±.005 | **.950**±.002 | **.909**±.006 |
| PM | **.885**±.005 | **.750**±.004 | **.950**±.002 | **.909**±.007 |
| Pair-only; matched events | | | | |
| Direct paint | .884±.006 | **.754**±.002 | .946±.007 | **.916**±.005 |
| Render+M3 | **.888**±.006 | **.754**±.003 | **.950**±.002 | .913±.006 |

is .680/.755 at 1× and .616/.630 at 8×. Both three-seed difference intervals include zero.

**Dynamics and phase scheduling.** With cycle and shape fixed, axis bias, sister constraints, exclusion and entry change F1 on 40 paired PSC programs by +0.37/+0.12 points at 1×/8× (two-test Holm $p \geq .058$). A single-phase timing study raises on-time detection by 8.58 points on 24 PSC scenes (block-bootstrap 95% interval [3.95,14.57]). The production two-phase policy (40 programs per dataset) raises first-attempt on-time matches from 140/1283 to 196/1310 on PSC and 295/868 to 348/872 on Fluo. With M3 and training budgets matched, tracking changes are +0.39/−0.16 points on PSC and −0.21/+0.24 on Fluo (1×/8×); none is significant after four-test Holm correction. Phase shifts may change later geometry.

**Independent label review.** A reviewer blinded to M3 decisions assessed 72 PSC events from 67 clips, eight per density-by-decision stratum. Division was visible in 23/24 kept, 16/24 re-timed and 10/24 dropped cases. Population-weighted rates are 95.9%, 66.5% and 41.2%. For retained events, the rate is 75.6% (approximate clip-clustered 95% interval [61.6,88.0]); the denominator includes uncertain/lost cases. The weighted absolute onset difference for 39 retained positives is 1.32 frames; human separation and M3 persistence criteria differ. Original clicks match training-ID triplets in 17/39 retained positives; daughter clicks precede label birth in 20 cases, and mother clicks occur at/after birth in two. Both daughter clicks carry the mother ID in 17 retained and 9/10 dropped positives; these timing/continuation disagreements are not an ancestry error rate. For eight preselected timing-conflict cases, the reviewer marked mothers at frame 0 and daughters at frame 16: 7/8 complete triplets match training IDs; one case has the same ID at both daughter points. This targeted check does not estimate corpus-wide accuracy; inter-rater reliability is untested.

**Uncertain supervision.** Excluding all triplets touching re-timed/dropped families, including descendants, changes F1 by −0.31/−0.26 PSC and +0.09/−0.15 Fluo points against count-matched random subsampling (1×/8×; five seeds; Holm $p \geq .374$). This policy also removes valid examples; it does not establish a benefit from masking.

### 4.5. Frozen-model transfer to new videos

Frozen Fluo models are evaluated on 36 LFCT fields (four cell lines [23]), without retraining or tuning. Shared exclusions from 53 fields are 11 with multiple-parent annotations, five integrity failures and one unavailable field. Exact mother–daughter event F1 uses shared Cellpose-SAM detections and centroid-in-mask assignment; it differs from CTC BC(0). Graph/R/RG means are .140/.130/.126 at 1× and .098/.138/.141 at 8×, averaging offsets, seeds, then fields. Paired field-bootstrap R/RG sparse gains over graph are 4.05/4.33 points (four-test Bonferroni intervals [2.38,5.25]/[2.90,5.83]); native intervals include zero. RG minus R is −0.34/+0.28 points; adjusted intervals include zero. Matched rendering/M3 trails painting by 0.41/0.54 points (both intervals below zero in a separate four-test family). Sparse R–graph gains are positive in all four cell lines (2.24–6.04 points); native signs vary. Intervals describe this field cohort; biological independence is unconfirmed.

With frozen R scores and continuity disabled, ILP exceeds greedy selection of the same child-set utility by 0.45/1.45 points (1×/8×; six-test Bonferroni intervals [0.21,0.75] and [0.82,2.17]). Continuity changes sparse R/RG F1 by −0.25/−0.29 points (adjusted intervals below zero); native intervals include zero. This is not a replication of [13].

**Computational cost.** PSC/Fluo job times sum to 5.41/5.40 hours for LoRA, 5.12/3.97 for generation with retries and 2.35/1.96 for readout, excluding label-editing overhead. Painting 200 clips takes 7.49/7.69 CPU minutes. Generation chunks run concurrently on 96 GB RTX PRO 6000 GPUs; per-job hardware and total elapsed time were not recorded.

## 5. DISCUSSION AND CONCLUSION

MitoSync jointly selects divisions and continuations using daughter-pair evidence and link scores. Component ablations support joint daughter scoring; fixed-score controls support joint decoding. Sparse LFCT gains support transfer beyond the development videos.

Synthetic supervision improves some fixed-tracker settings, but painting remains competitive and timing gains do not consistently improve tracking. The identity review shows that detector consistency alone cannot establish lineage accuracy. Joint association provides the clearest tracking benefit; synthetic supervision should be judged by training utility and lineage accuracy.

## 6. COMPLIANCE WITH ETHICAL STANDARDS

We used public Cell Tracking Challenge [21] and LFCT [23] data. No new human or animal data were collected, so ethical approval was not required. The authors have no relevant financial or non-financial interests to disclose.

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The review package specifies all inputs, training settings, checkpoint hashes, LFCT scoring rules, exclusions, per-view counts and error decomposition. Code, generated datasets and evaluation records will be released at `https://github.com/CHEN-Yunshu/MitoSync-ISBI2027`.

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